DATA ENGINEERING · TECHNICAL TEST IILUMA

Full Run Evidence

Job postings pipeline — raw → staging → core on PostgreSQL

785,741

rows read into **raw**

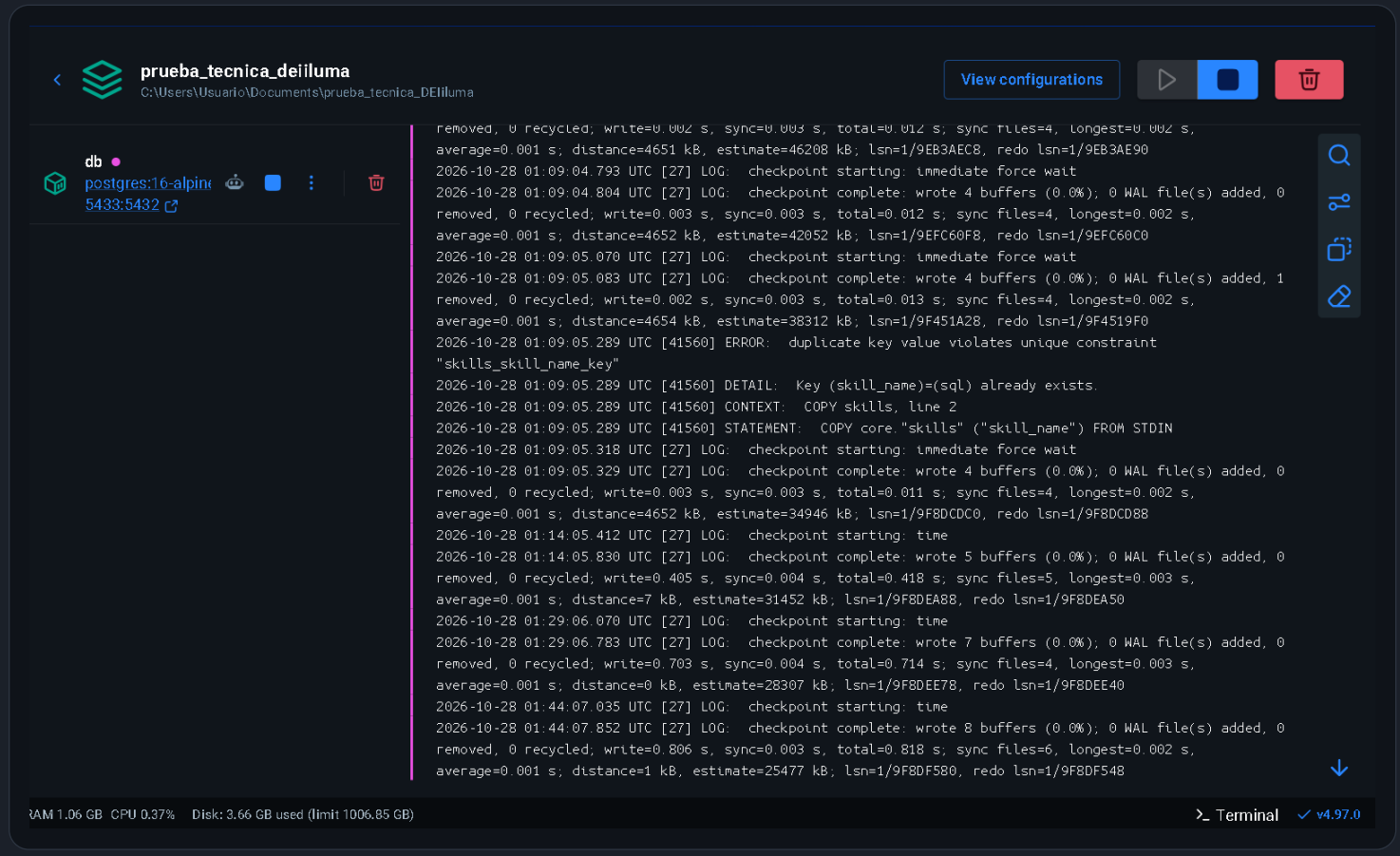
785,237

jobs loaded into **core**

01 · THE DATABASE

PostgreSQL 16 running in Docker

This slide shows where the pipeline ran: the **db** service from **docker-compose.yml**, image **postgres:16-alpine**, host port 5433 mapped to the container's 5432.



The ERROR in the log is expected

The **skills_skill_name_key** duplicate-key error is the integration test proving that a duplicate key makes **TableLoader** roll back.

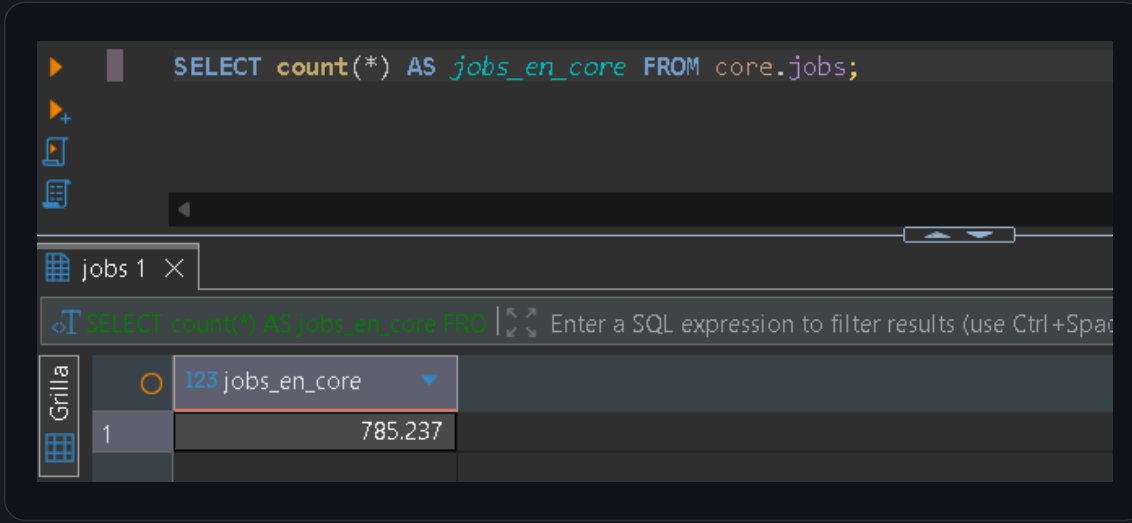
docs/docker_postgres_running.png

02 · THE RESULT

785,237 jobs in core

This slide reconciles the count. Every posting that reached **core.jobs** is a distinct row of **raw** that passed validation; the arithmetic below accounts for each one dropped.

Distinct postings in raw	785,640
Salary rate but no amount	− 402
Missing title	− 1
Rows in core.jobs	785,237



03 · DATA CLEANING, ON THE SAME POSTING

Two fixes, each shown on one row

This slide proves the cleaning on real rows. The posting in **raw** and in **core** is joined by **row_hash = source_row_hash**, so each pair is the same row in both layers: left column as the file has it, right column after cleaning.

Mojibake repaired

docs/case_mojibake.png

caso	raw_title	core_title
mojibake	Consultant Datascientist - Risques de CrÃ©dit et Climatiques H/F	Consultant Datascientist - Risques de Crédit et Climatiques H/F

raw keeps **CrÃ©dit** exactly as the file has it; **staging** repairs it to **Crédit**, because the whole text is UTF-8 that was read as LATIN1.

Repeated spaces collapsed

docs/fix_data_spaces.png

espacios	(Global Oil Gas) Data Scientist - Time Series Forecasting	(Global Oil Gas) Data Scientist - Time Series Forecasting

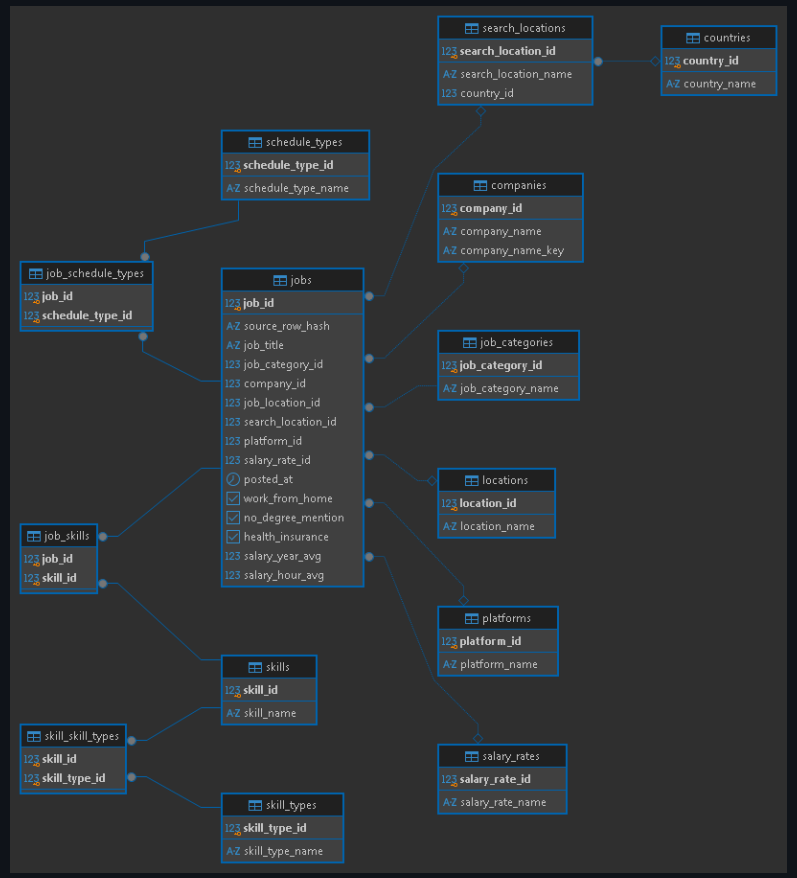
The double space in **Scientist - Time** is kept in **raw** and collapsed to one in **core**.

04 · THE TABLES OF CORE

One fact table, ten dimensions, three bridges

Jobs holds the posting and foreign keys only; every repeated value lives in its own dimension, and every list-valued field became a bridge table.

Fact	jobs
Dimensions	companies · locations · search_locations · countries · platforms · job_categories · salary_rates · schedule_types · skills · skill_types
Bridges	job_schedule_types · job_skills · skill_skill_types

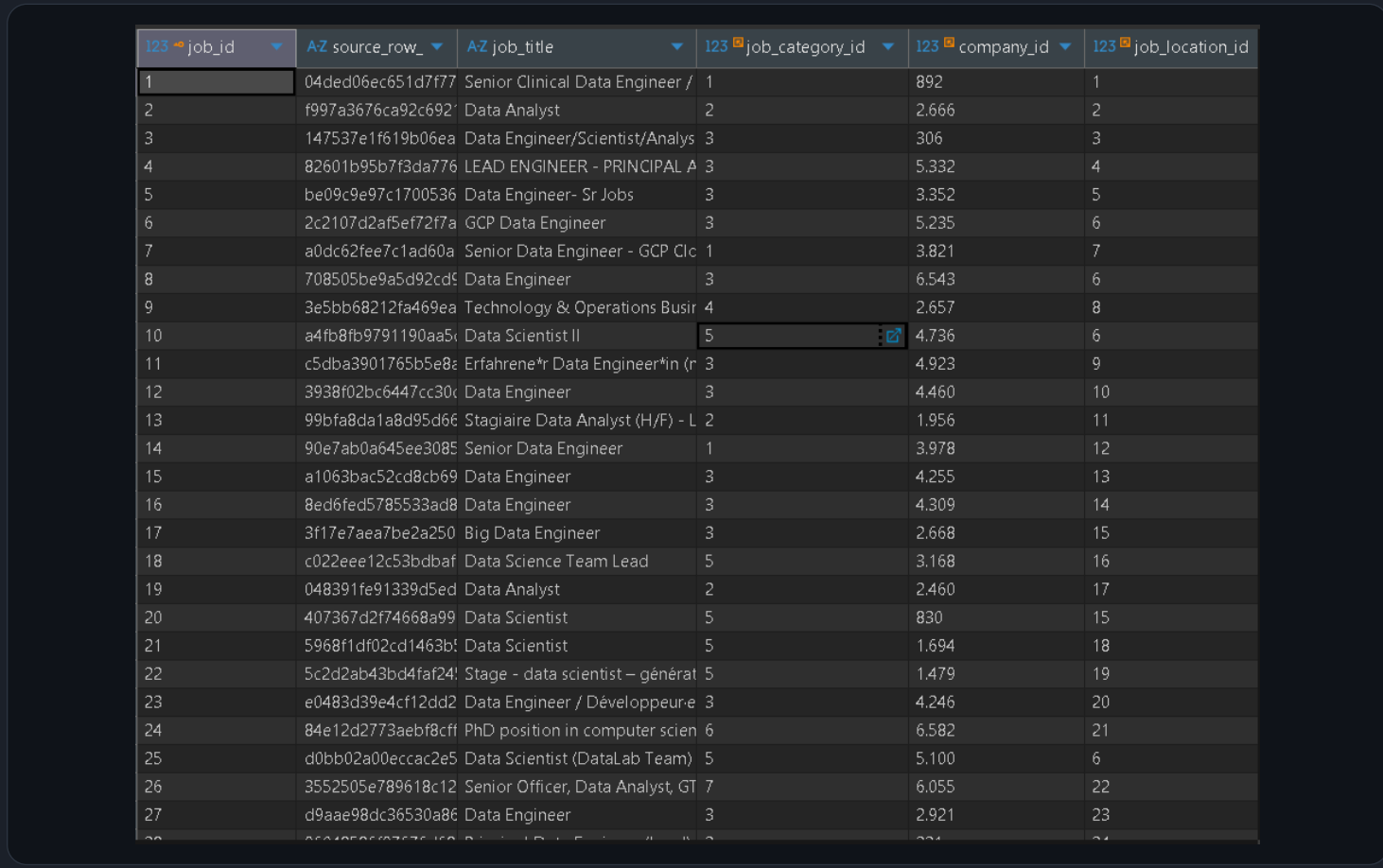


05 · CORE.JOBS

The fact table: the posting and its keys

This image shows the first rows of **jobs**. It holds only the posting itself and foreign keys; nothing descriptive is repeated here.

- **source_row_hash** is the 64-character SHA-256 of the raw row.
- Ids are assigned by the database, not by the loader.

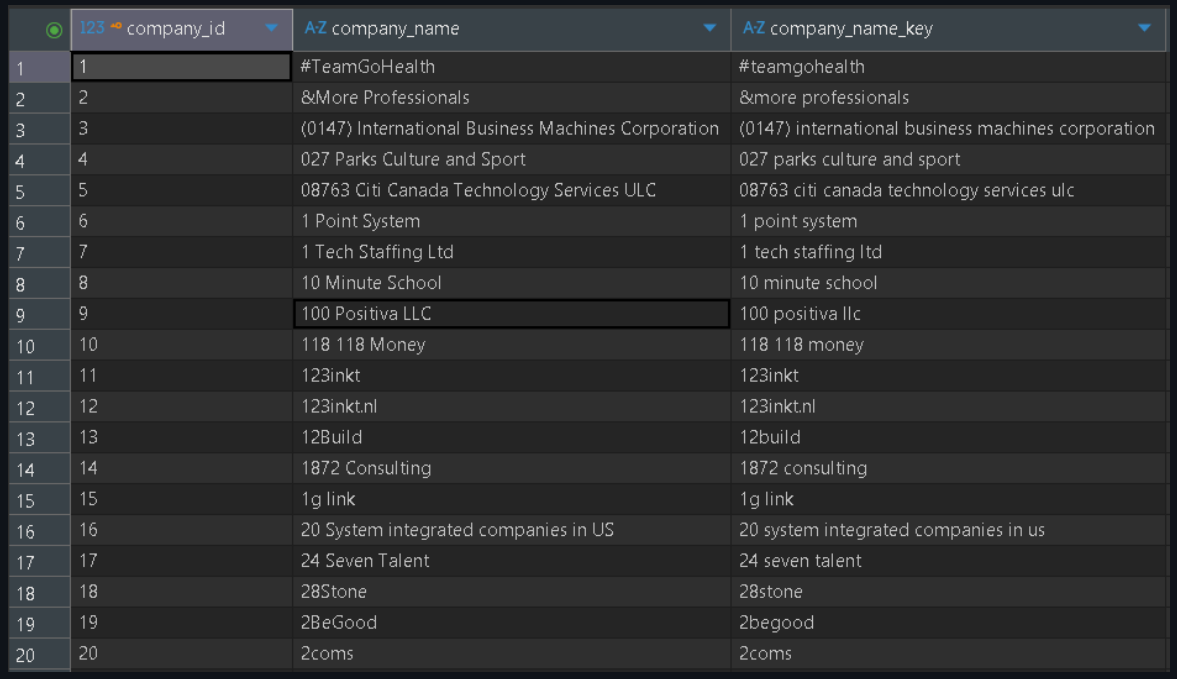


job_id	source_row_	job_title	job_category_id	company_id	job_location_id
1	04ded06ec651d7f77	Senior Clinical Data Engineer /	1	892	1
2	f997a3676ca92c692	Data Analyst	2	2.666	2
3	147537e1f619b06ea	Data Engineer/Scientist/Analys	3	306	3
4	82601b95b7f3da776	LEAD ENGINEER - PRINCIPAL A	3	5.332	4
5	be09c9e97c1700536	Data Engineer- Sr Jobs	3	3.352	5
6	2c2107d2af5ef72f7a	GCP Data Engineer	3	5.235	6
7	a0dc62fee7c1ad60a	Senior Data Engineer - GCP Clo	1	3.821	7
8	708505be9a5d92cd9	Data Engineer	3	6.543	6
9	3e5bb68212fa469ea	Technology & Operations Busir	4	2.657	8
10	a4fb8fb9791190aa5	Data Scientist II	5	4.736	6
11	c5dba3901765b5e8a	Erfahrene*r Data Engineer*in (r	3	4.923	9
12	3938f02bc6447cc30	Data Engineer	3	4.460	10
13	99bfa8da1a8d95d66	Stagiaire Data Analyst (H/F) - L	2	1.956	11
14	90e7ab0a645ee3085	Senior Data Engineer	1	3.978	12
15	a1063bac52cd8cb69	Data Engineer	3	4.255	13
16	8ed6fed5785533ad8	Data Engineer	3	4.309	14
17	3f17e7aea7be2a250	Big Data Engineer	3	2.668	15
18	c022eee12c53bdbaf	Data Science Team Lead	5	3.168	16
19	048391fe91339d5ed	Data Analyst	2	2.460	17
20	407367d2f74668a99	Data Scientist	5	830	15
21	5968f1df02cd1463b5	Data Scientist	5	1.694	18
22	5c2d2ab43bd4faf245	Stage - data scientist – générat	5	1.479	19
23	e0483d39e4cf12dd2	Data Engineer / Développeur·e	3	4.246	20
24	84e12d2773aebf8cff	PhD position in computer scien	6	6.582	21
25	d0bb02a00eccac2e5	Data Scientist (DataLab Team)	5	5.100	6
26	3552505e789618c12	Senior Officer, Data Analyst, GT	7	6.055	22
27	d9aae98dc36530a86	Data Engineer	3	2.921	23

06 · DIMENSIONS · COMPANIES AND LOCATIONS

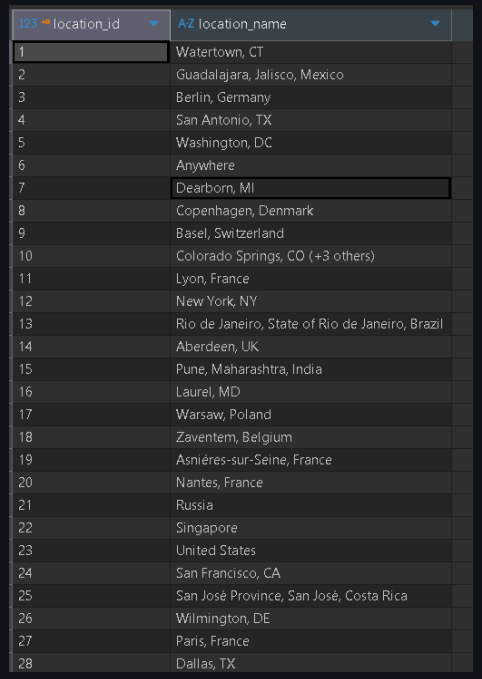
Grouped by key, stored as written

This images shows the two dimensions that come straight from free text. Companies are deduplicated through a lowercase key; locations are kept verbatim because the files own values are valid.



	company_id	company_name	company_name_key
1	1	#TeamGoHealth	#teamgohealth
2	2	&More Professionals	&more professionals
3	3	(0147) International Business Machines Corporation	(0147) international business machines corporation
4	4	027 Parks Culture and Sport	027 parks culture and sport
5	5	08763 Citi Canada Technology Services ULC	08763 citi canada technology services ulc
6	6	1 Point System	1 point system
7	7	1 Tech Staffing Ltd	1 tech staffing ltd
8	8	10 Minute School	10 minute school
9	9	100 Positiva LLC	100 positiva llc
10	10	118 118 Money	118 118 money
11	11	123inkt	123inkt
12	12	123inkt.nl	123inkt.nl
13	13	12Build	12build
14	14	1872 Consulting	1872 consulting
15	15	1g link	1g link
16	16	20 System integrated companies in US	20 system integrated companies in us
17	17	24 Seven Talent	24 seven talent
18	18	28Stone	28stone
19	19	2BeGood	2begood
20	20	2coms	2coms

companies — **company_name_key** is the lowercase form used to group spellings; **company_name** keeps the most frequent one. Ids follow the alphabetical order of the key because each chunk is grouped by key before inserting.



	location_id	location_name
1	1	Watertown, CT
2	2	Guadalajara, Jalisco, Mexico
3	3	Berlin, Germany
4	4	San Antonio, TX
5	5	Washington, DC
6	6	Anywhere
7	7	Dearborn, MI
8	8	Copenhagen, Denmark
9	9	Basel, Switzerland
10	10	Colorado Springs, CO (+3 others)
11	11	Lyon, France
12	12	New York, NY
13	13	Rio de Janeiro, State of Rio de Janeiro, Brazil
14	14	Aberdeen, UK
15	15	Pune, Maharashtra, India
16	16	Laurel, MD
17	17	Warsaw, Poland
18	18	Zaventem, Belgium
19	19	Asnières-sur-Seine, France
20	20	Nantes, France
21	21	Russia
22	22	Singapore
23	23	United States
24	24	San Francisco, CA
25	25	San José Province, San José, Costa Rica
26	26	Wilmington, DE
27	27	Paris, France
28	28	Dallas, TX

locations — stored as written in the file, so **Anywhere** and **Colorado Springs, CO (+3 others)** are valid values of the source.

07 · DIMENSIONS · SEARCH_LOCATIONS AND COUNTRIES

The country hangs from the search, not the job

Here slide shows the two-level geography. A job points to a search location; the search location points to a country. That is the design decision, and it keeps the file's own labelling even when it is surprising.

search_location_id	search_location_name	country_id
1	Texas, United States	1
2	Mexico	2
3	Germany	3
4	Sudan	4
5	Georgia	1
6	Florida, United States	1
7	Romania	5
8	Denmark	6
9	New York, United States	1
10	Switzerland	7
11	France	8
12	Brazil	9
13	United Kingdom	10
14	India	11
15	Poland	12
16	Belgium	13
17	Russia	14
18	Serbia	15
19	Singapore	16
20	Costa Rica	17

search_locations — **Georgia** points to the United States because that is how the file labels it.

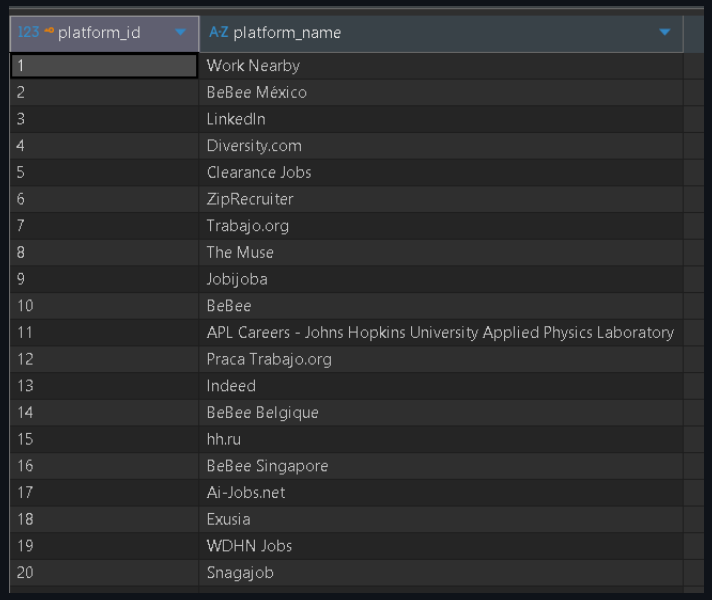
country_id	country_name
1	United States
2	Mexico
3	Germany
4	Sudan
5	Romania
6	Denmark
7	Switzerland
8	France
9	Brazil
10	United Kingdom
11	India
12	Poland
13	Belgium
14	Russia
15	Serbia
16	Singapore
17	Costa Rica
18	Ireland
19	Italy
20	Malaysia

countries — 160 countries, numbered in order of first appearance in **raw**.

08 · DIMENSIONS · PLATFORMS, JOB_CATEGORIES, SALARY_RATES

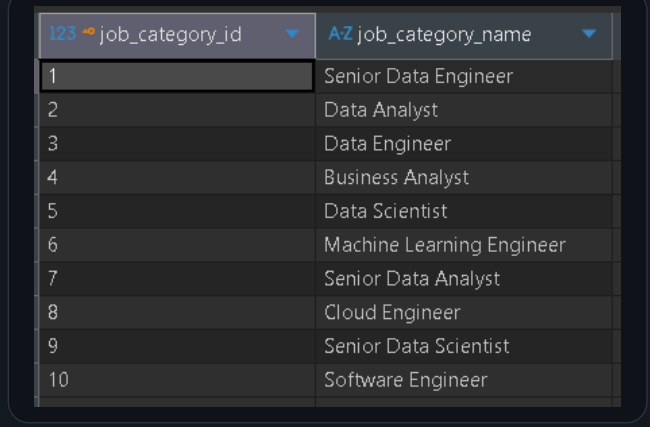
Small lookups, each with one rule

This slide groups the three short dimensions. Each one applies exactly one normalization rule, and the rule is stated under the capture.



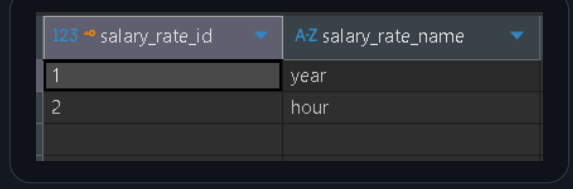
platform_id	platform_name
1	Work Nearby
2	BeBee México
3	LinkedIn
4	Diversity.com
5	Clearance Jobs
6	ZipRecruiter
7	Trabajo.org
8	The Muse
9	Jobijoba
10	BeBee
11	APL Careers - Johns Hopkins University Applied Physics Laboratory
12	Praca Trabajo.org
13	Indeed
14	BeBee Belgique
15	hh.ru
16	BeBee Singapore
17	Ai-Jobs.net
18	Exusia
19	WDHN Jobs
20	Snagajob

platforms — the **via** prefix is gone. **BeBee**, **BeBee México** and **BeBee Singapore** stay separate, because merging them would be a guess.



job_category_id	job_category_name
1	Senior Data Engineer
2	Data Analyst
3	Data Engineer
4	Business Analyst
5	Data Scientist
6	Machine Learning Engineer
7	Senior Data Analyst
8	Cloud Engineer
9	Senior Data Scientist
10	Software Engineer

job_categories — the standardized title (**job_title_short**): only 10 values.



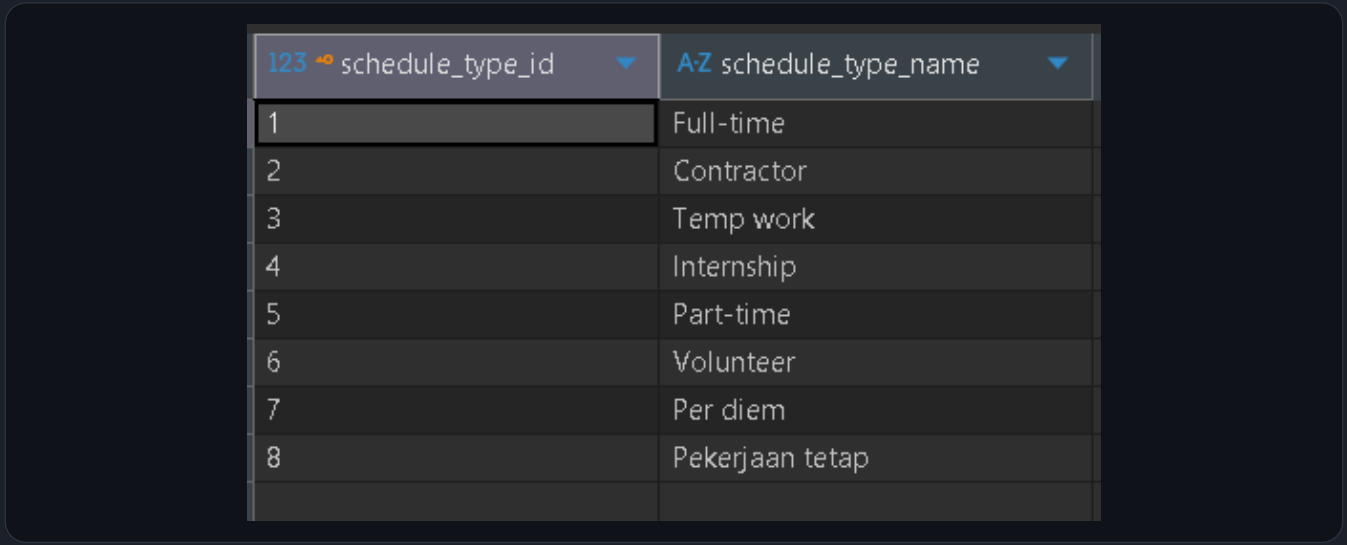
salary_rate_id	salary_rate_name
1	year
2	hour

salary_rates — only **year** and **hour**. **month**, **week** and **day** appeared only in the 402 postings without an amount, which are rejected.

09 · SCHEDULE_TYPES AND JOB_SCHEDULE_TYPES

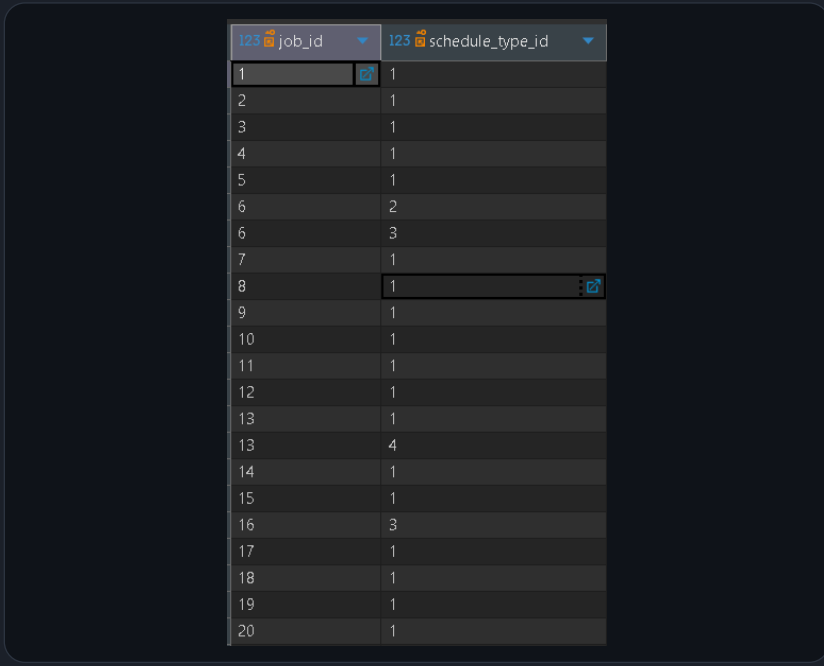
Atomic values, joined through a bridge

This shows how a combined value like **“Full-time and Part-time”** is split: the dimension holds eight atomic types, and the bridge table gives a job as many rows as it has types.



schedule_type_id	schedule_type_name
1	Full-time
2	Contractor
3	Temp work
4	Internship
5	Part-time
6	Volunteer
7	Per diem
8	Pekerjaan tetap

schedule_types — eight atomic values, never combinations. **Kontraktor** was translated to **Contractor** (id 2); **Pekerjaan tetap** is the other non-English value of the source and is kept.

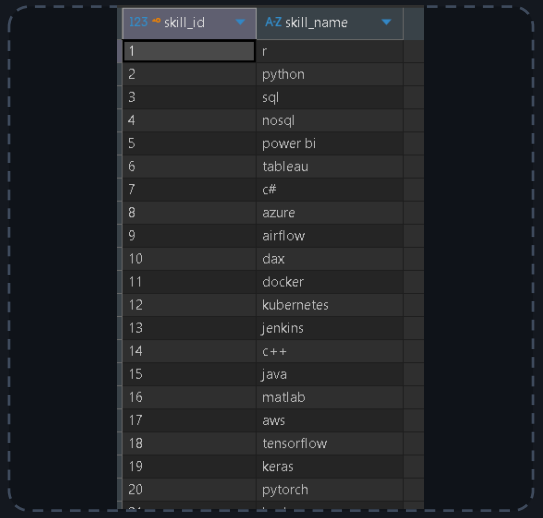


job_id	schedule_type_id
1	1
2	1
3	1
4	1
5	1
6	2
6	3
7	1
8	1
9	1
10	1
11	1
12	1
13	1
13	4
14	1
15	1
16	3
17	1
18	1
19	1
20	1

job_schedule_types — the bridge that replaces the combined string: job 6 has two rows (types 2 and 3) and job 13 has types 1 and 4.

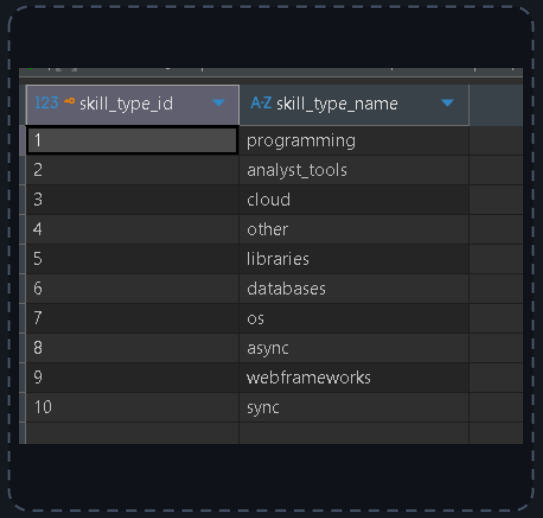
Skills: two dimensions, two bridges

This slide covers the skills model. A skill can belong to several types and a job can list several skills, so both relations are bridge tables with composite primary keys.



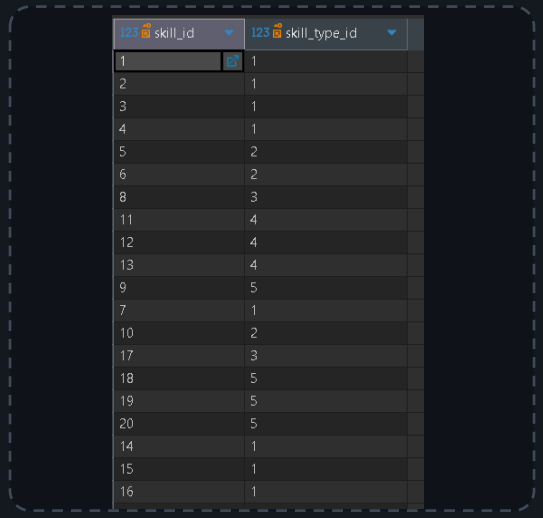
skill_id	skill_name
1	r
2	python
3	sql
4	nosql
5	power bi
6	tableau
7	c#
8	azure
9	airflow
10	dax
11	docker
12	kubernetes
13	jenkins
14	c++
15	java
16	matlab
17	aws
18	tensorflow
19	keras
20	pytorch

skills — 252 skills, lowercase and without repeats, numbered by first appearance.



skill_type_id	skill_type_name
1	programming
2	analyst_tools
3	cloud
4	other
5	libraries
6	databases
7	os
8	async
9	webframeworks
10	sync

skill_types — the 10 skill types of the file (**programming**, **analyst_tools**, **cloud**, **libraries**, **databases**…).



skill_id	skill_type_id
1	1
2	1
3	1
4	1
5	2
6	2
8	3
11	4
12	4
13	4
9	5
7	1
10	2
17	3
18	5
19	5
20	5
14	1
15	1
16	1

skill_skill_types — bridge between skills and their types, with a composite primary key.



job_id	skill_id
2	1
2	2
2	3
2	4
2	5
2	6
3	2
3	3
3	7
3	8
3	9
3	10
3	11
3	12
3	13
4	2
4	14
4	15
4	16
4	17
4	18
4	19
4	20

job_skills — starts at job 2 because job 1 has no skills (117K postings have none); jobs 2, 3 and 4 share skill 2.

11 · THE MANY - TO- MANY RELATI ON WORKI NG

One job, eleven rows, one skill in two types

This is the functional proof of the bridges. Job 10 (**Data Scientist II**) is joined through **job_skills**, **skills**, **skill_skill_types** and **skill_types**.

Why a single type column would fail

mongodb appears twice, once as a **database** and once as a **programming** skill. Only a bridge with a composite key can hold both.

job_id	job_title	skill_name	skill_type_name
10	Data Scientist II	azure	cloud
10	Data Scientist II	excel	analyst_tools
10	Data Scientist II	mongodb	databases
10	Data Scientist II	mongodb	programming
10	Data Scientist II	pandas	libraries
10	Data Scientist II	python	programming
10	Data Scientist II	r	programming
10	Data Scientist II	spark	libraries
10	Data Scientist II	sql	programming
10	Data Scientist II	sql server	databases
10	Data Scientist II	windows	os

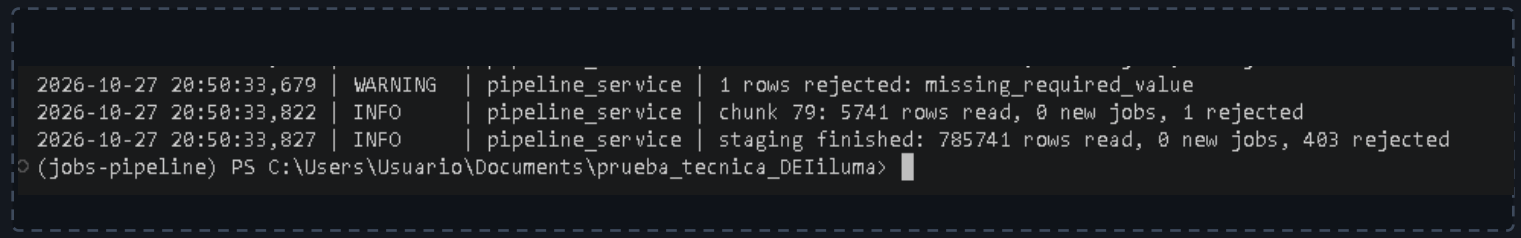
12 · IDEMPOTENCY

Run it twice: same 785,741 read, 0 new jobs

Here it shows the second run of **staging**. It reads the full file again and loads nothing new, because every row is already in **core** under its hash.

0
new jobs loaded

403
rows rejected again



The 403 rejected rows are logged again with their reason (**missing_required_value** for the posting without a title) because they stay in **raw**.